Database modelling

MBUA 512

Class 2 – where we're going

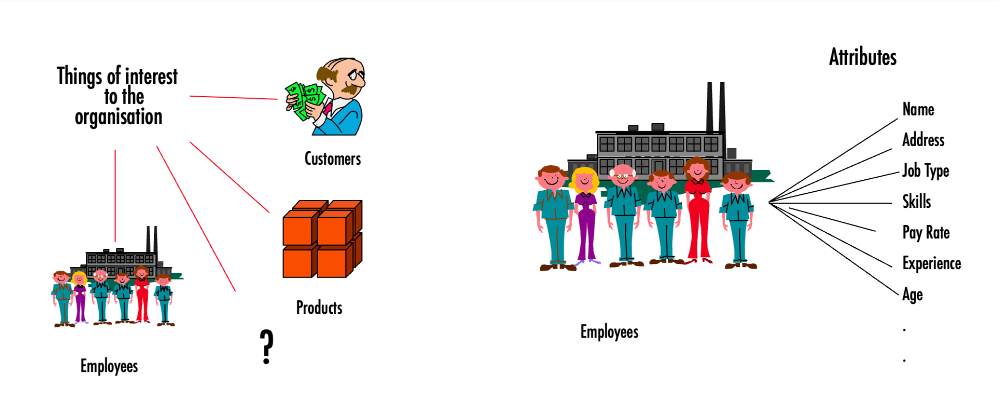
Database modelling, with Michael Winikoff

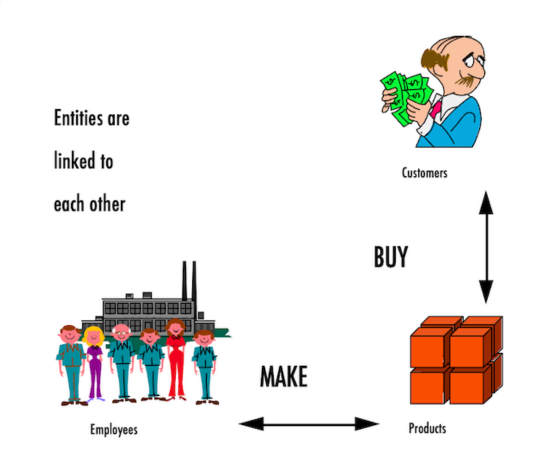
1. Data modelling using diagrams
2. Translating diagrams to relational databases, and the role of *keys*
3. Anomalies, and avoiding them by *normalisation*

Based on slides by Professor Markus Luczak-Roesch.

Recap from last week

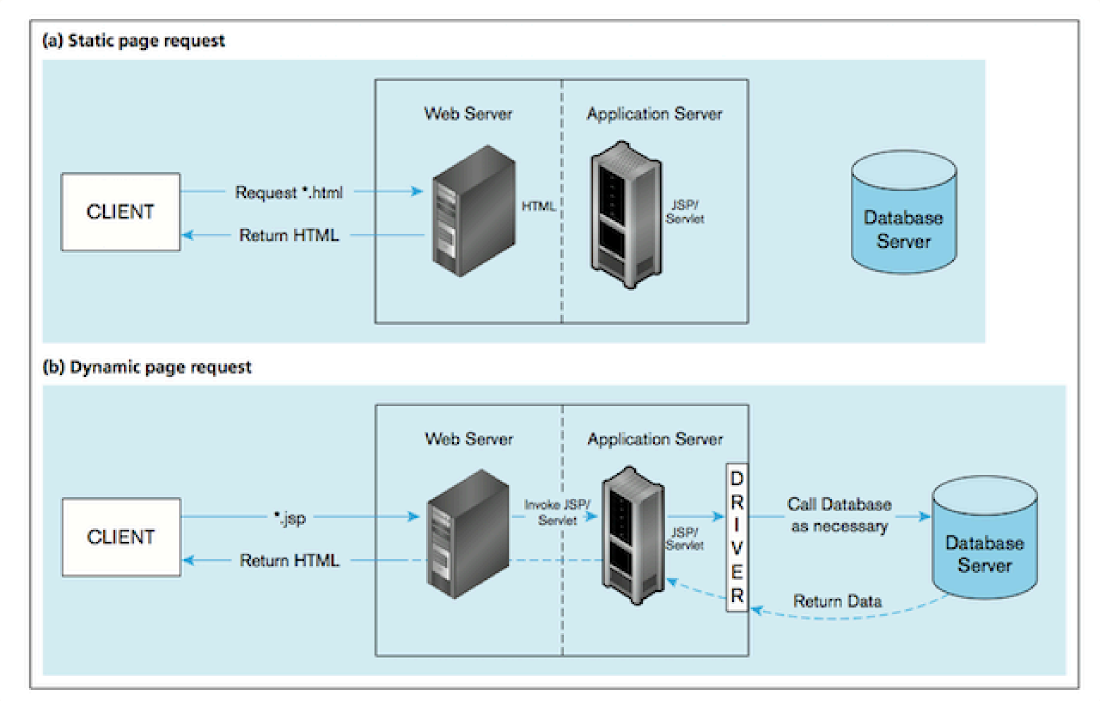
- What is data?
  - Types of data (qualitative/quantitative, structured/unstructured)
  - Why is data important ... to organisations? ... to business analysts?
- What is a DB management system and what is a *relational* database? (technology)
- Concepts: **entity**, **attributes**, **relationships** (modelling data)

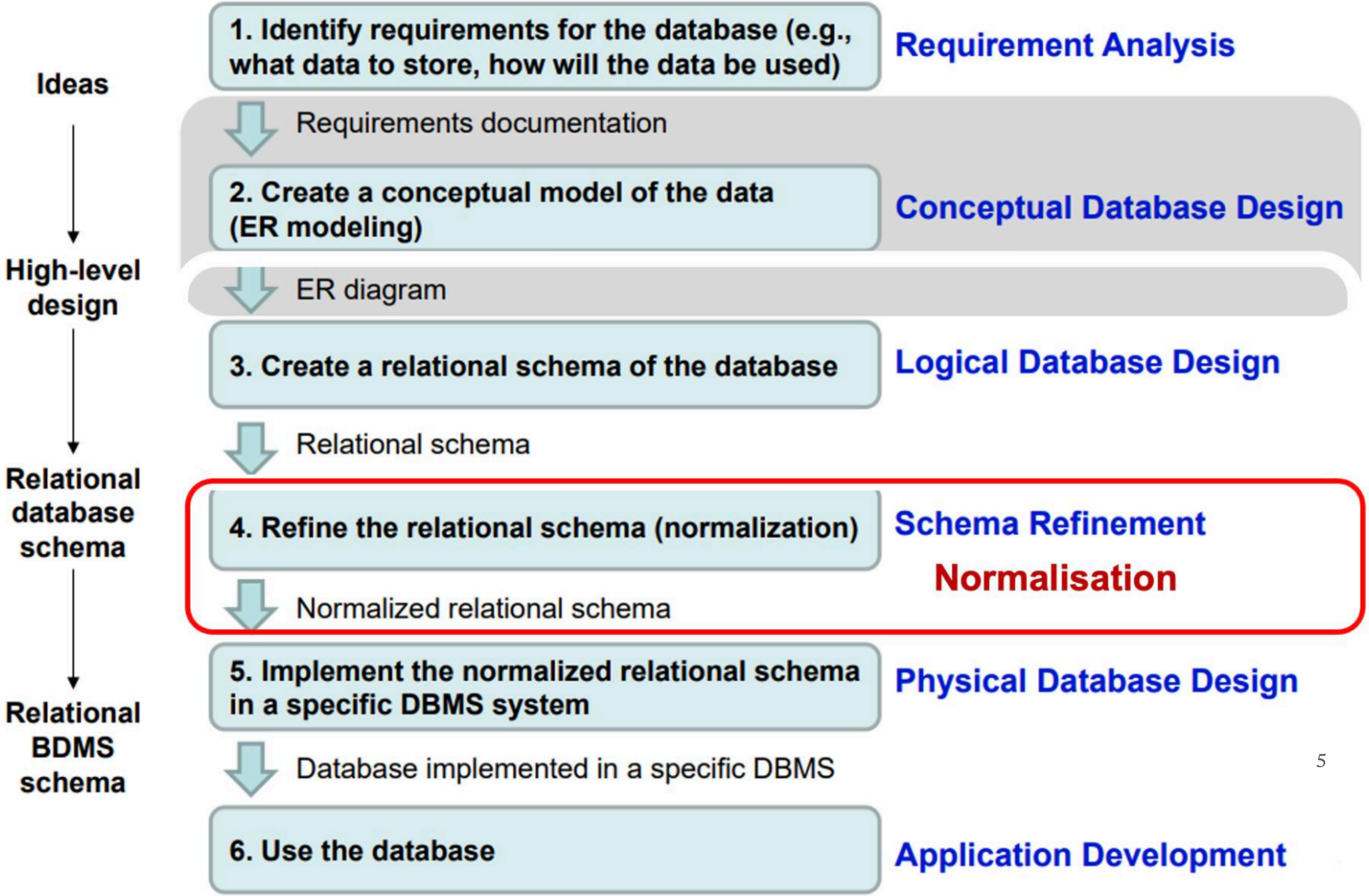




Big picture

Roles: DB analysis, designers, developers, front-end app analysts and developers, DB admin



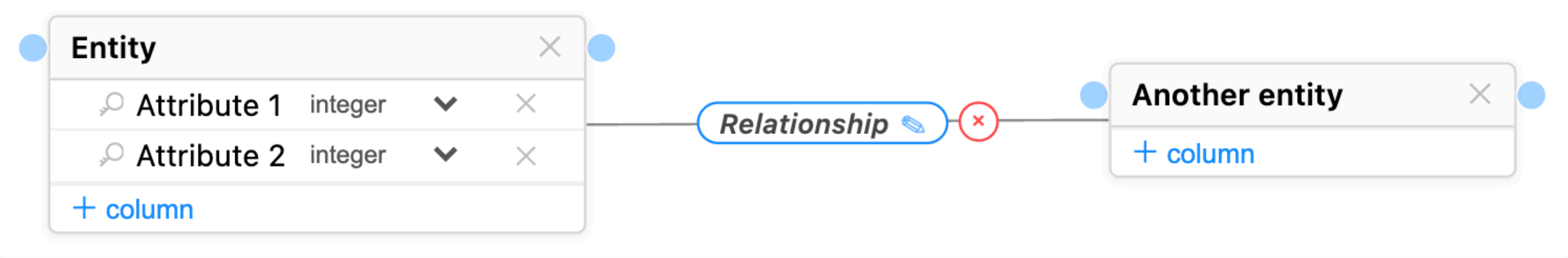

Ideas
High-level design
Relational database schema
Relational BDMS schema
1. Identify requirements for the database (e.g., what data to store, how will the data be used)
Requirement Analysis
Requirements documentation
2. Create a conceptual model of the data (ER modeling)
Conceptual Database Design
ER diagram
3. Create a relational schema of the database
Logical Database Design
Relational schema
4. Refine the relational schema (normalization)
Schema Refinement
Normalisation
Normalized relational schema
5. Implement the normalized relational schema in a specific DBMS system
Physical Database Design
Database implemented in a specific DBMS
6. Use the database
Application Development

Data modelling using diagrams

1. Data modelling using diagrams

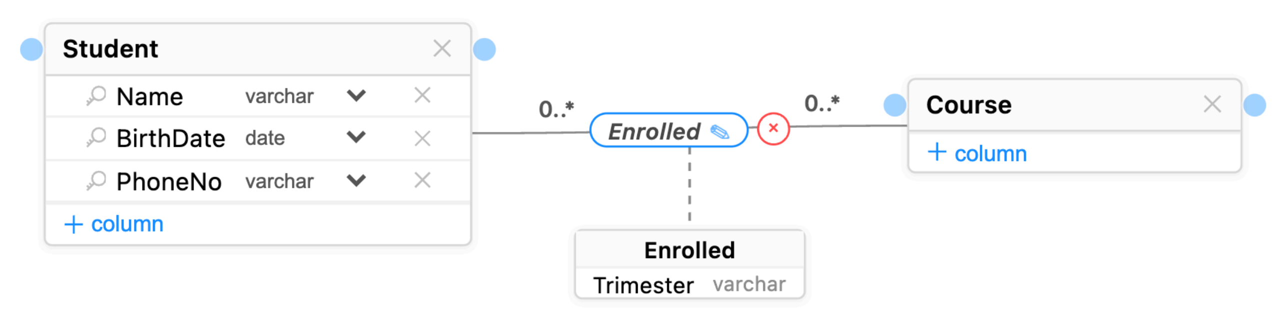
Model *conceptual* model in terms of entities, attributes, relationships using an **ER (Entity-Relationship) diagram**:

- Entity = rectangle with entity name at the top
- Attribute = in the entity rectangle, below the entity name, and with a type (e.g. integer, varchar, date)
- Relationship = link between entities (with optional label)



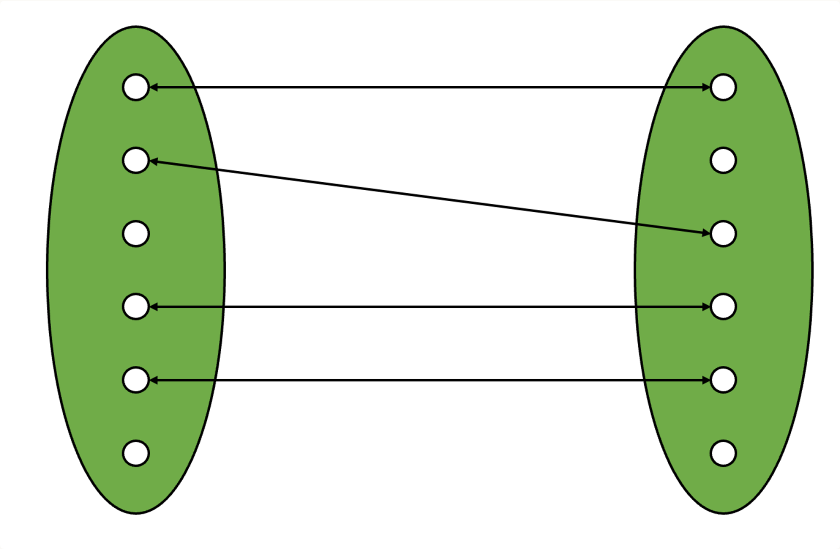
Relationships ...

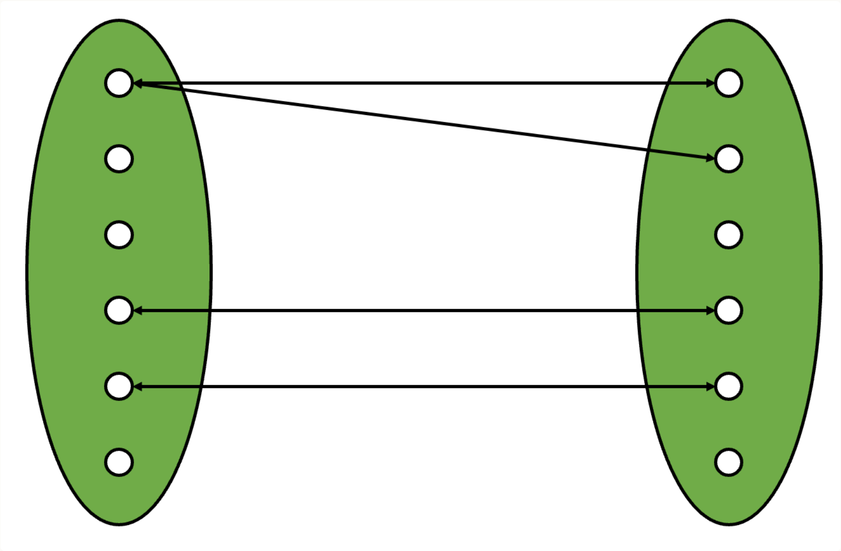
- A relationship links one or more entities, and can have attributes (shown in a dash-linked rectangle)
- Relationships have **degree** and **cardinality**:
  - Degree: number of participating entity **types** in a relationship
    - i.e. how many entities does the relationship link in the diagram (typically 2, sometimes 1)
  - Cardinality: number of entity **instances** (next slide)



Cardinality – number of *instances* of entities

Left: one-to-one (e.g. bus and bus driver); Right: one-to-many (e.g. bus and passenger)

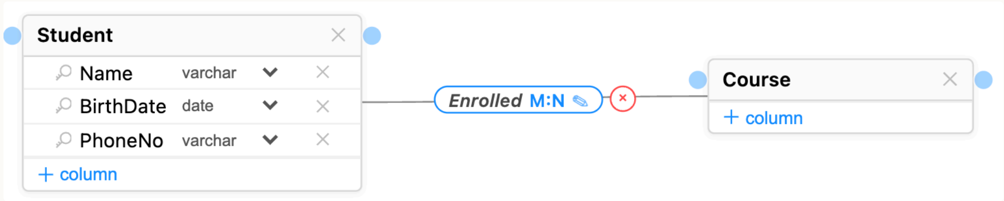




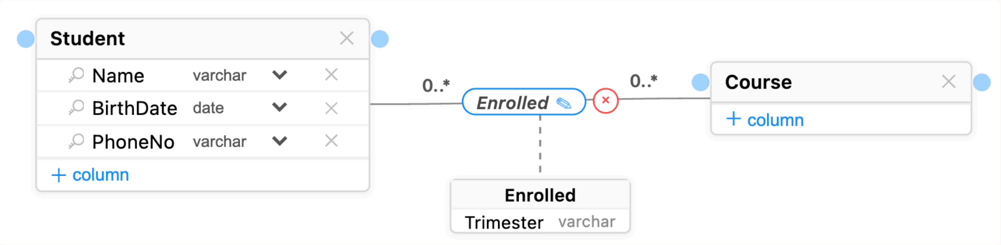
Notation for cardinality

There are a number of different notations that can be used: pick one and be consistent!

- Chen: uses 1:1, 1:N, N:1, M:N labels



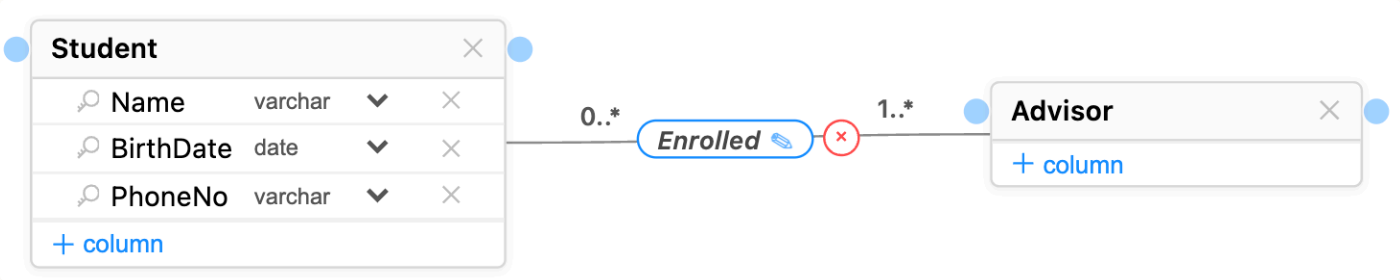
- UML: uses *low..high* at each end of the relationship with `*` indicating "many", so 1:N would be `1` at one end (short for `1..1`) and `0..*` at the other end



Notation for cardinality

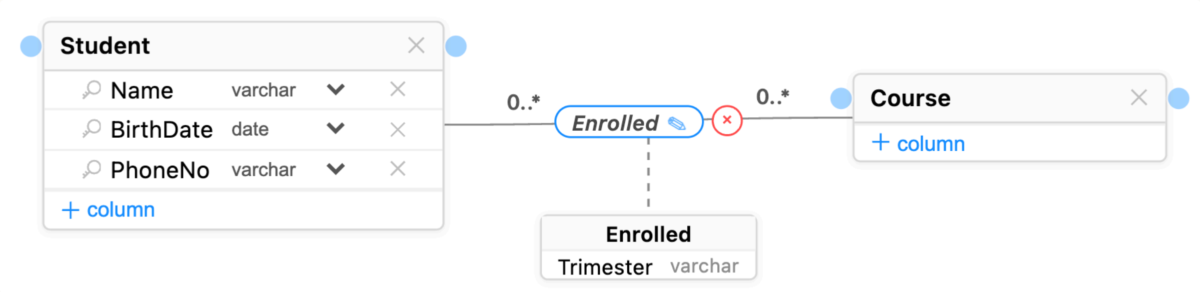
The Chen notation is simpler, but doesn't allow distinction between 0 (optional) and 1 (mandatory) to be modelled

Example: a student must have at least one advisor



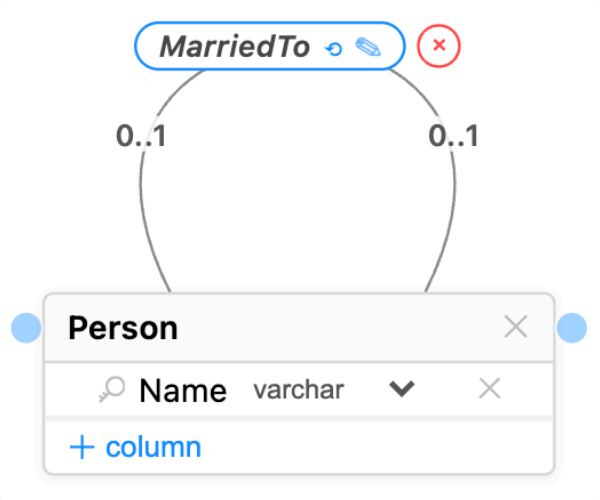
Examples of degree and cardinality (1/2)

- Enrolled is Binary degree: two entity types, student and course
- Enrolled is N-N cardinality: a student can enrol in many courses, and a course can have many students



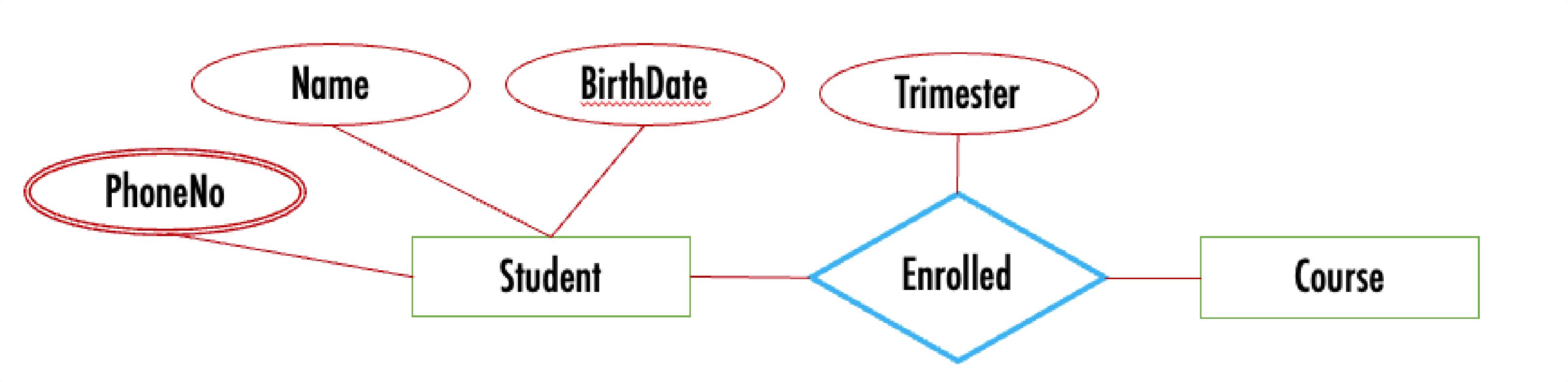
Examples of degree and cardinality (2/2)

- MarriedTo is Unary degree: only one type of entity is involved: a Person
- MarriedTo is 1-1 cardinality: each individual person is married to one other person



Alternative notation for ER diagrams

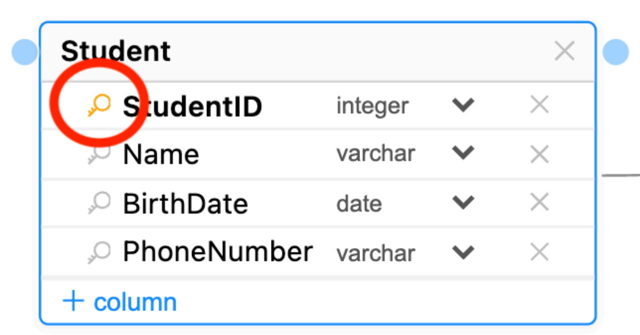
- Attribute: oval (with double line for multiple values, and dashed line for derived attributes [not shown below])
- Relationship: Diamond

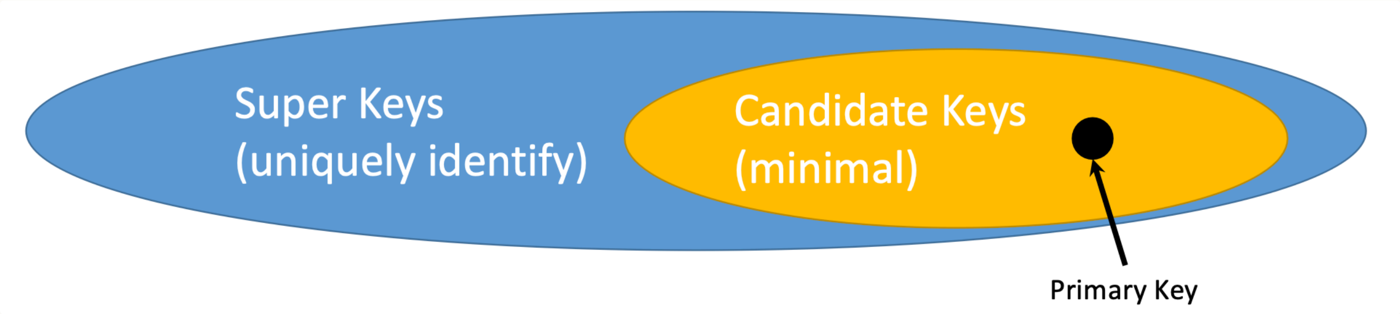


> We won't be using this notation in this course

Keys

- A **key** is one or more attributes that must be unique to each entity instance, i.e. no two instances can have the same value for their key attribute(s) (so e.g. name is a bad choice)
    - A *super key* is a set of attributes that can be a key
    - A *candidate key* is a *minimal* super key (no subset of it is a super key)
    - A *primary key* is a chosen candidate key
- Notation: key attribute indicated with key icon or PK





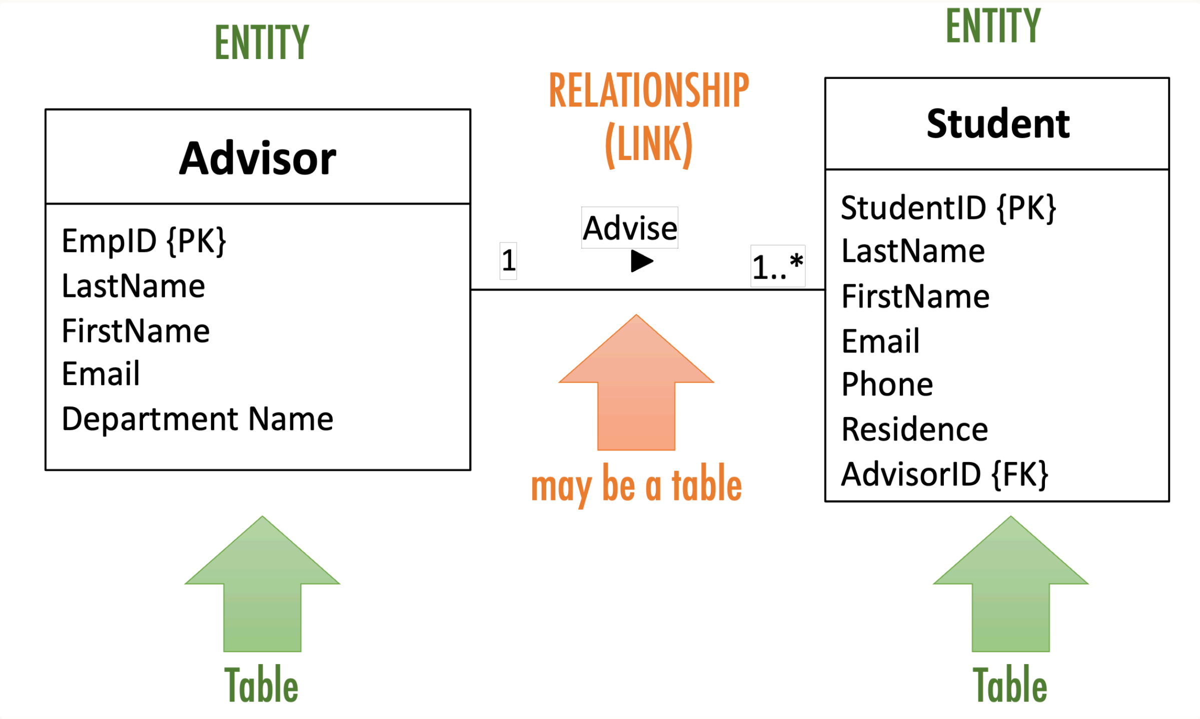
Tips for modelling

- Iterative!
- Need to focus on what is important, rather than capture everything
- Careful with notation: link entities and associations with their attributes, an association needs to be linked to one or more entities
- Naming:
  - Brief, but clear
  - Unique - e.g. don't use "name" for both "student name" and "employee name"
  - Entities and attributes: singular ("student")
  - Relationships: verbs or verb phrases ("inspects", "manages", "belongsTo")

Translating diagrams to relational databases

2. Translating diagrams to relational databases

- Each entity becomes a table (relation) containing the entity's attributes
- Each relationship is mapped to either a table or additional attributes, depending on the cardinalities



Mapping a 1-N relationship as an additional attribute

Advisor (left) is in a 1-N relationship ("Advise") with Student (right)

Implement by adding to each student their advisor's ID (since there is at most 1)

EmpID	LastName
2463	
1942	

StudentID	LastName	Advisor
4511	...	2463
8463	...	2463
9617	...	1942

Mapping an N-N relationship as a table

Advisor (left) is in an N-N relationship ("Advise") with Student (right)

Implement by creating a separate table for the relationship

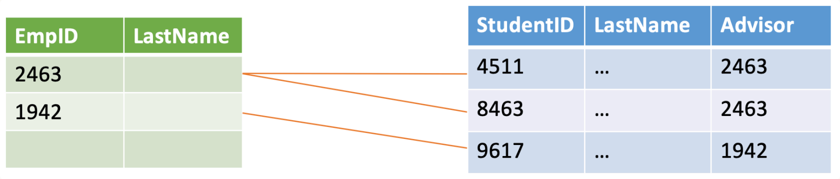
EmpID	LastName
2463	
1942	

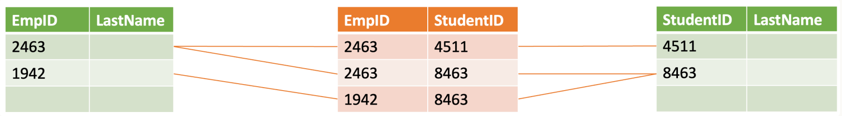
EmpID	StudentID
2463	4511
2463	8463
1942	8463

StudentID	LastName
4511	
8463	

Foreign keys

- A *key* is the attribute(s) that can be used to identify an entity instance (e.g. StudentID)
- Each table (relation) will have column(s) for its keys (e.g. the Student table has a column StudentID)
- A *foreign key* is when a table has a column that contains keys for *another table* (e.g. Student table contains the ID of the student's advisor)
  - in UML notation indicated with `{FK}`





Anomalies and Normalisation

3. Anomalies and Normalisation

Errors or inconsistencies occurring when updating tables with redundant data:

- insertion anomaly example: require data about different entities to be entered at the same time in a single relation – cannot track customer address separately from order
- deletion anomaly example: deletion of a row results in loss of multiple facts – deleting the last order loses the customer's contact details!
- modification anomaly example: need to modify multiple rows to update a fact consistently – update address requires updating every order

Example on the next slides.

AD CAMPAIGN MIX

AdCampaignID	AdCampaign Name	StartDate	Duration	Campaign MgrID	Campaign MgrName	ModusID	Media	Range	Budget Pctg
111	SummerFun20	6.6.2020	12 days	CM100	Roberta	1	TV	Local	50%
111	SummerFun20	6.6.2020	12 days	CM100	Roberta	2	TV	National	50%
222	SummerZing20	6.8.2020	30 days	CM101	Sue	1	TV	Local	60%
222	SummerZing20	6.8.2020	30 days	CM101	Sue	3	Radio	Local	30%
222	SummerZing20	6.8.2020	30 days	CM101	Sue	5	Print	Local	10%
333	FallBall20	6.9.2020	12 days	CM102	John	3	Radio	Local	80%
333	FallBall20	6.9.2020	12 days	CM102	John	4	Radio	National	20%
444	AutmnStyle20	6.9.2020	5 days	CM103	Nancy	6	Print	National	100%
555	AutmnColors20	6.9.2020	3 days	CM100	Roberta	3	Radio	Local	100%

AD CAMPAIGN MIX

AdCampaignID	AdCampaign Name	StartDate	Duration	Campaign MgrID	Campaign MgrName	ModusID	Media	Range	Budget Pctg
111	SummerFun20	6.6.2020	12 days	CM100	Roberta	1	TV	Local	50%
111	SummerFun20	6.6.2020	12 days	CM100	Roberta	2	TV	National	50%
222	SummerZing20	6.8.2020	30 days	CM101	Sue	1	TV	Local	60%
222	SummerZing20	6.8.2020	30 days	CM101	Sue	3	Radio	Local	30%
222	SummerZing20	6.8.2020	30 days	CM101	Sue	5	Print	Local	10%
333	FallBall20	6.9.2020	12 days	CM102	John	3	Radio	Local	80%
333	FallBall20	6.9.2020	12 days	CM102	John	4	Radio	National	20%
444	AutmnStyle20	6.9.2020	5 days	CM103	Nancy	6	Print	National	100%
555	AutmnColors20	6.9.2020	3 days	CM100	Roberta	3	Radio	Local	100%

The media and range values for campaign modus 1 repeated twice

The name of the campaign manager CM100 repeated three times

The name, start date, and duration of the campaign 222 repeated three times

AD CAMPAIGN MIX

AdCampaignID	AdCampaign Name	StartDate	Duration	Campaign MgrID	Campaign MgrName	ModusID	Media	Range	Budget Pctg
111	SummerFun20	6.6.2020	12 days	CM100	Roberta	1	TV	Local	50%
111	SummerFun20	6.6.2020	12 days	CM100	Roberta	2	TV	National	50%
222	SummerZing20	6.8.2020	**30 days**	CM101	Sue	1	TV	Local	60%
222	SummerZing20	6.8.2020	**30 days**	CM101	Sue	3	Radio	Local	30%
222	SummerZing20	6.8.2020	**30 days**	CM101	Sue	5	Print	Local	10%
333	FallBall20	6.9.2020	12 days	CM102	John	3	Radio	Local	80%
333	FallBall20	6.9.2020	12 days	CM102	John	4	Radio	National	20%
444	**AutmnStyle20**	**6.9.2020**	**5 days**	CM103	Nancy	6	Print	National	100%
555	AutmnColors20	6.9.2020	3 days	CM100	Roberta	3	Radio	Local	100%
????	????	????	????	????	????	**7**	**Internet**	**National**	**????**

Deletion Anomaly Example:
Cannot delete campaign 444 without also deleting all the data about campaign manager CM103 and campaign modus 6.

Modification Anomaly Example:
To change the duration of the campaign 222 from 30 to 45 days, three records have to be modified.

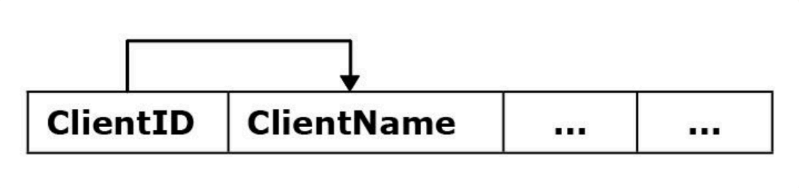
Insertion Anomaly Example:
Cannot insert new campaign modus 7 without inserting an actual campaign using the new modus 7.

Solution: *normalise* by splitting up the table

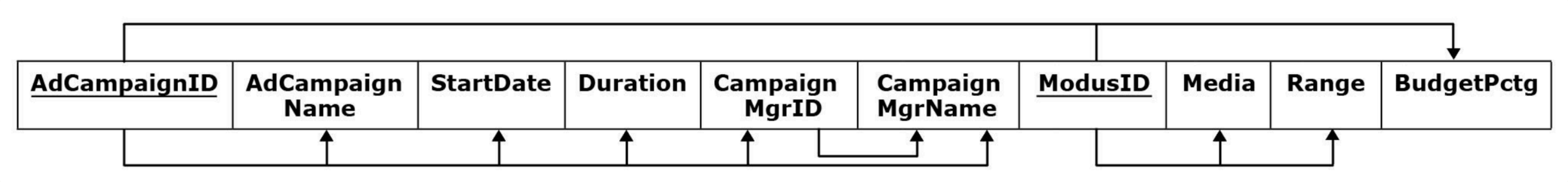
How to do this? By analysing *functional dependencies*.

Functional dependency - occurs when the value of one (or more) column(s) in each record of a relation uniquely determines the value of another column in that same record of the relation

Example: ClientID → ClientName

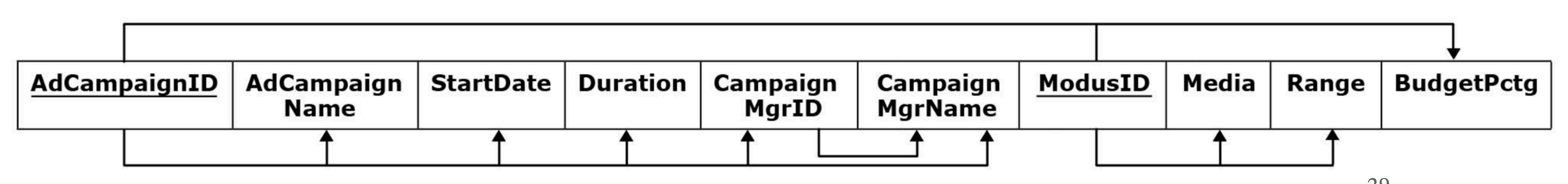


Dependencies for the ad campaign DB

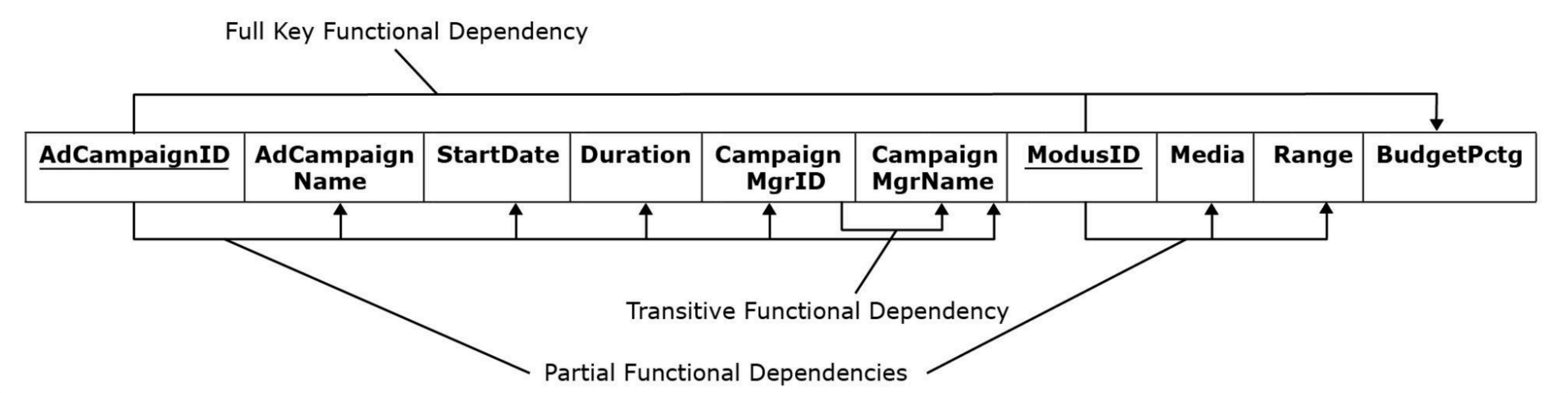


Types of functional dependencies

- ❌ **Partial**: a column depends on a *component* of a composite primary key; e.g. ModusID → Media, Range
- ✅ **Full**: the primary key functionally determines the column, and no separate component of it partially determines the same column
- ❌ **Transitive**: nonkey columns functionally determine other nonkey columns; e.g. CampaignMgrID → CampaignMgrName



Dependencies for the ad campaign DB



Normalisation - 1NF

- First Normal Form (1NF): no multi-value entries
- Normalise by splitting records

VET CLINIC CLIENT

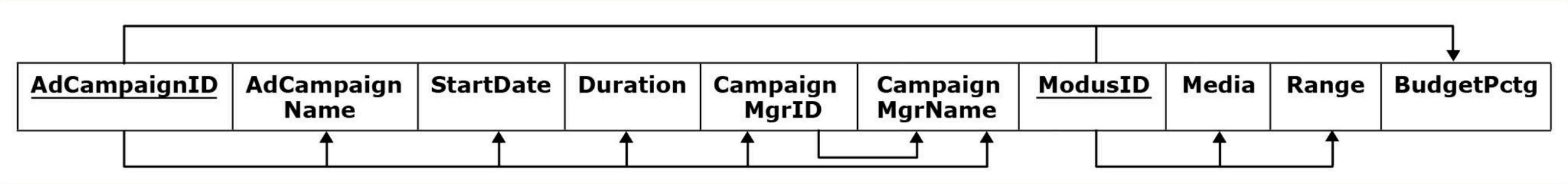
ClientID	ClientName	PetNo	PetName	PetType
111	Lisa	1	Tofu	Dog
222	Lydia	1 2 3	Fluffy JoJo Ziggy	Dog Bird Snake
333	Jane	1 2	Fluffy Cleo	Cat Cat

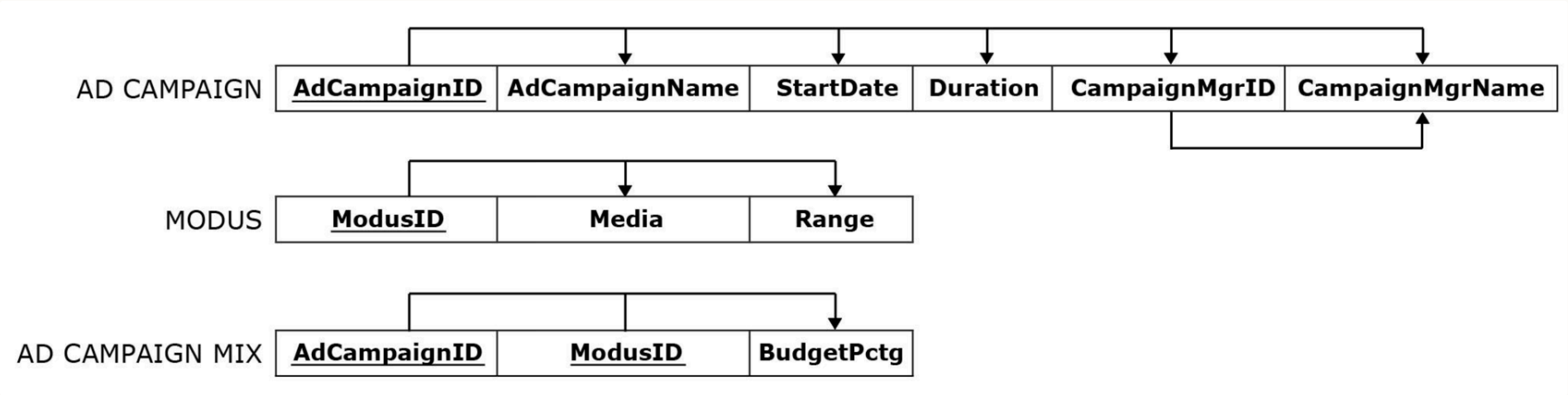
VET CLINIC CLIENT

ClientID	ClientName	PetNo	PetName	PetType
111	Lisa	1	Tofu	Dog
222	Lydia	1	Fluffy	Dog
222	Lydia	2	JoJo	Bird
222	Lydia	3	Ziggy	Snake
333	Jane	1	Fluffy	Cat
333	Jane	2	Cleo	Cat

Normalisation - 2NF

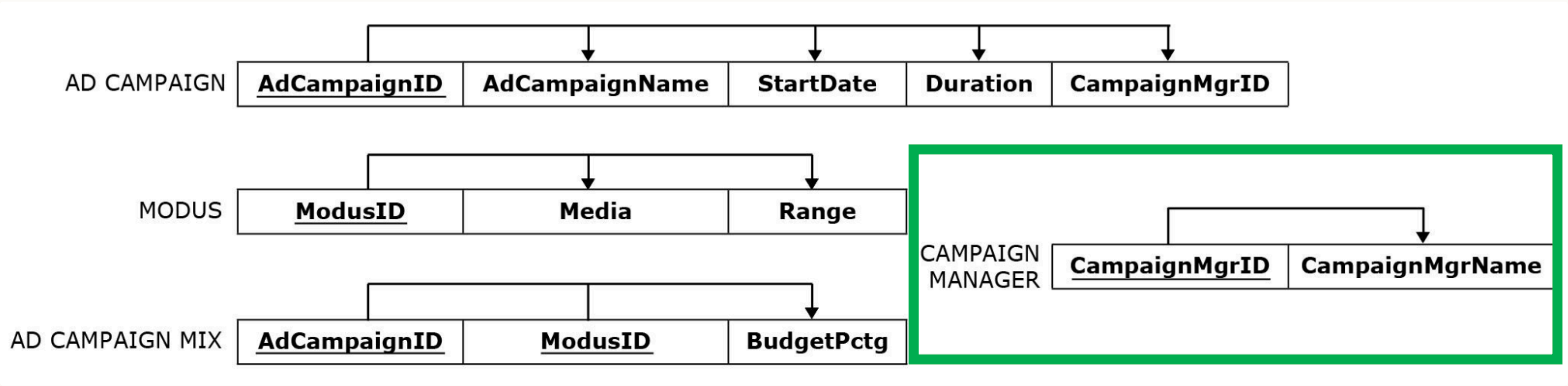
- Second Normal Form (2NF): a table is in 2NF if it is in 1NF and contains no **partial** functional dependencies – one way to ensure this is a single-column primary key
- Normalise by splitting out things that depend on part of the key into a separate table





Normalisation - 3NF

- Third Normal Form (3NF): A table is in 3NF if it is in 2NF and if it does not contain transitive functional dependencies
- Normalise by splitting out things that depend on nonkey columns into a separate table



Result on the next slide.

AD CAMPAIGN

AdCampaignID	AdCampaignName	StartDate	Duration	CampaignMgrID
111	SummerFun20	6.6.2020	12 days	CM100
222	SummerZing20	6.8.2020	45 days	CM101
333	FallBall20	6.9.2020	12 days	CM102
555	AutmnColors20	6.9.2020	3 days	CM100

Modification Anomaly Resolved:
Only one record modified.

CAMPAIGN MANAGER

CampaignMgrID	CampaignMgrName
CM100	Roberta
CM101	Sue
CM102	John
CM103	Nancy

Deletion Anomaly Resolved:
Campaign 444 deleted but all the data about the campaign manager CM103 and the campaign modus 6 remain in the database.

MODUS

ModusID	Media	Range
1	TV	Local
2	TV	National
3	Radio	Local
4	Radio	National
5	Print	Local
6	Print	National
7	Internet	National

Insertion Anomaly Resolved:
New campaign modus 7 inserted without inserting an actual campaign using the new modus 7.

AD CAMPAIGN MIX

AdCampaignID	ModusID	BudgetPctg
111	1	50%
111	2	50%
222	1	60%
222	3	30%
222	5	10%
333	3	80%
333	4	20%
555	3	100%

Normalisation Summary

- Having too many things (columns) in a table can cause anomalies when adding/deleting/changing records
- This occurs when there are partial or transitive functional dependencies
- Solution is to normalise to 3NF:
    1. Identify functional dependencies
    2. Split out columns to remove partial and transitive functional dependencies

This is probably the trickiest database concept we cover!

Good news: typically a database that results from a design process (identifying entities, attributes, relationships) is in 3NF

Today

1. Data modelling using ER diagrams
2. Translating diagrams to relational databases, and the role of *keys*
3. Anomalies, and avoiding them by *normalisation*

Next: activities in part 2 today – learn by doing.

Next week, building, modifying and querying relational databases with SQL ("es-queue-el", or "sequel").

Activities

Activity – modelling your organisation

~ *20 minutes, in groups* Select one of the organisations below and then:

1. Identify some *entities* and *attributes* (not necessarily for all entities)
2. Identify at least one *relationship*
3. Sketch an ER diagram showing the entities, attributes, and relationship(s) identified, as well as cardinalities
4. Map your ER diagram to a collection of tables - which relationships are mapped to tables and which to additional attributes?

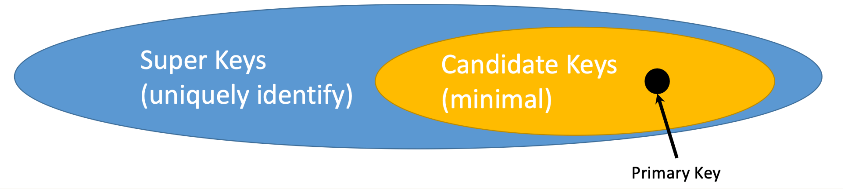
Organisations: the tax office (IRD), a large store (e.g. The Warehouse), VUW, a local café.

Activity – identify candidate keys

~ 10 minutes, in groups

In the table below what would be possible choices for a primary key? Why?

Flight Number	Airline	Day	Time	Pilot ID	Pilot Name	Plane model	Plane manufacturer
NZ123	Air NZ	10 Aug 2018	10am	PL8715	Pat Smith	Airbus310	AirBus
NZ456	Air NZ	11 Aug 2018	11am	PL8716	Yi Zhang	Airbus320	AirBus
YA789	Yugoslav Air	17 Aug 2018	10:30am	PL9781	Jim Belich	Boeing737	Boeing
NZ123	Air NZ	12 Aug 2018	10am	PL9781	Jim Belich	Boeing737	Boeing
NZ124	Air NZ	13 Aug 2018	11pm	PL9781	Jim Belich	Airbus310	Airbus
YA890	Yugoslav Air	17 Aug 2018	4pm	PL9781	Jim Belich	Airbus310	Airbus
…	…	…	…	…	…	…	…



Activity – normalisation

~20 minutes, in groups

- Identify the dependencies in the table on the next slide – assume the primary key is Flight Number and Day.
- How would you transform it into 3NF? Identify functional dependencies, classify them, and split to remove partial and transitive dependencies.

Assumptions: a flight number always flies at the same time; a flight number does not always use the same plane model. What else do you need to assume?

Flight Number	Airline	Day	Time	Pilot ID	Pilot Name	Plane model	Plane manufacturer
NZ123	Air NZ	10 Aug 2018	10am	PL8715	Pat Smith	Airbus310	AirBus
NZ456	Air NZ	11 Aug 2018	11am	PL8716	Yi Zhang	Airbus320	AirBus
YA789	Yugoslav Air	17 Aug 2018	10:30am	PL9781	Jim Belich	Boeing737	Boeing
NZ123	Air NZ	12 Aug 2018	10am	PL9781	Jim Belich	Boeing737	Boeing
NZ124	Air NZ	13 Aug 2018	11pm	PL9781	Jim Belich	Airbus310	Airbus
YA890	Yugoslav Air	17 Aug 2018	4pm	PL9781	Jim Belich	Airbus310	Airbus
…	…	…	…	…	…	…	…